

Grade 12 lessons

Pre-calculus and calculus

IN THIS BOOK

1. Unit circle
2. Limits
3. Derivatives
4. Integrals
5. Counting and probability

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 12 Kahoot and Blooket sets, and one stop on the Grade 12 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Unit circle

THE BIG IDEA

The unit circle is a circle of radius 1 centered at the origin. For any angle, cosine is the x value on the circle and sine is the y value.

HOW TO DO IT

1. Know the key values: $\sin 0^\circ = 0$, $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 90^\circ = 1$.
2. Cosine takes the same values in the opposite order: $\cos 0^\circ = 1$ down to $\cos 90^\circ = 0$.
3. Radians: a full turn is 2π , so $180^\circ = \pi$. Multiply degrees by $\frac{\pi}{180}$ to convert.
4. Signs by quadrant: all positive in I, only sine in II, only tangent in III, only cosine in IV.
5. Two identities to know: $\sin^2 \theta + \cos^2 \theta = 1$ and $\tan \theta = \sin \theta \div \cos \theta$.

WORKED EXAMPLES

Convert 60° to radians.

$$60 \times \frac{\pi}{180} = \frac{\pi}{3}.$$

What is $\cos 180^\circ$?

At 180° the point on the circle is $(-1, 0)$. Cosine is the x value: -1 .

WATCH OUT

Cosine is x and sine is y. Do not swap them. And π radians is 180° , not 360° .

WORDS TO KNOW

radian

an angle measure where a full turn is 2π

identity

an equation that is true for every angle

Limits

THE BIG IDEA

A limit is the value a function gets closer and closer to as x approaches some number. It is the idea that calculus is built on.

HOW TO DO IT

1. First try plugging the number in. For polynomials that gives the limit.
2. If you get $\frac{0}{0}$, factor the top and bottom, cancel, then plug in again.
3. As x goes to infinity, a number divided by x goes to 0.
4. For a fraction of polynomials as x goes to infinity: if the top and bottom have the same highest power, the limit is the ratio of their leading numbers.

WORKED EXAMPLES

Find the limit of $(x^2 - 9) \div (x - 3)$ as $x \rightarrow 3$.

Plugging in gives $\frac{0}{0}$. Factor: $(x + 3)(x - 3) \div (x - 3) = x + 3$. Now plug in: $3 + 3 = 6$.

Find the limit of $(6x^2 + 1) \div (2x^2 + 5x)$ as $x \rightarrow \infty$.

Both have x^2 as the highest power. $6 \div 2 = 3$.

WATCH OUT

Getting $\frac{0}{0}$ does not mean the limit is 0 or that it does not exist. It means simplify first.

WORDS TO KNOW

limit

the value a function approaches

infinity

growing without any bound

Derivatives

THE BIG IDEA

The derivative measures how fast something is changing. On a graph it is the slope of the curve at a point.

HOW TO DO IT

1. Power rule: for x^n , bring the power down in front and lower the power by one. x^3 becomes $3x^2$.
2. A number in front stays and multiplies: $4x^5$ becomes $20x^4$.
3. Take the derivative of each term separately. A plain number becomes 0.
4. To find the slope at a point, plug the x value into the derivative.
5. Know these: $\sin x \rightarrow \cos x$, $\cos x \rightarrow -\sin x$, $e^x \rightarrow e^x$, $\ln x \rightarrow \frac{1}{x}$.

WORKED EXAMPLES

$f(x) = 3x^2 + 5x - 7$. Find $f'(x)$.

$3x^2$ becomes $6x$. $5x$ becomes 5 . -7 becomes 0 . So $f'(x) = 6x + 5$.

What is the slope of that curve at $x = 2$?

$f'(2) = 6 \times 2 + 5 = 17$.

WATCH OUT

Multiply by the power first, then lower the power. And the derivative of a plain number is 0, not the number itself.

WORDS TO KNOW

derivative

the rate of change of a function

tangent line

a line that just touches the curve at one point

Integrals

THE BIG IDEA

Integrating is the reverse of differentiating. An indefinite integral gives a function. A definite integral gives a number: the area under the curve.

HOW TO DO IT

1. Power rule backwards: raise the power by one, then divide by the new power. x^2 becomes $\frac{x^3}{3}$.
2. For an indefinite integral, add $+ C$ at the end.
3. For a definite integral from a to b : find the antiderivative, plug in b , plug in a , and subtract.
4. The integral of a constant k from a to b is a rectangle: $k \times (b - a)$.
5. Know these: $\cos x \rightarrow \sin x$, $\sin x \rightarrow -\cos x$, $e^x \rightarrow e^x$, $\frac{1}{x} \rightarrow \ln|x|$.

WORKED EXAMPLES

$\int 6x^2 dx = ?$

Raise the power to 3 and divide by 3:
 $6 \div 3 = 2$. Answer: $2x^3 + C$.

$\int \text{from } 0 \text{ to } 3 \text{ of } 2x dx = ?$

The antiderivative is x^2 . At 3 it is 9. At 0 it is 0. $9 - 0 = 9$.

WATCH OUT

Do not forget $+ C$ on an indefinite integral. For a definite integral it is the top value minus the bottom value, in that order.

WORDS TO KNOW

antiderivative

a function whose derivative is the one you started with

definite integral

an integral between two limits

Counting and probability

THE BIG IDEA

Counting rules tell you how many ways something can happen without listing them all. Probability then compares the ways you want with all the ways.

HOW TO DO IT

1. $n!$ (n factorial) means $n \times (n - 1) \times \dots \times 1$. It counts the ways to arrange n things.
2. Permutations: order matters. Picking 3 from 5 in order is $5 \times 4 \times 3$.
3. Combinations: order does not matter. Divide the permutations by the number of orders of the group.
4. Independent events: multiply the probabilities.
5. Expected value: multiply each prize by its probability and add.

WORKED EXAMPLES

How many ways can 3 of 5 people stand in a line?

$$5 \times 4 \times 3 = 60.$$

How many teams of 3 can be chosen from 5 people?

Start with 60 ordered picks. Each team of 3 can be ordered $3! = 6$ ways.
 $60 \div 6 = 10$.

WATCH OUT

Ask first: does order matter? A line or a code: yes, use permutations. A team or a hand of cards: no, use combinations.

WORDS TO KNOW

factorial

the product of a number and every whole number below it

permutation

an ordered choice

combination

a choice where order does not matter